

RAJALAKSHMI ENGINEERING COLLEGE

Experiment 4: EDA - Data Inspection and Analysis

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AIM

To inspect and analyze a loan dataset using previews, schema information, descriptive statistics, categorical-value checks, type conversion, and statistical measures.

SOURCE

EXP4/EXP4.ipynb

PYTHON CODE - CELL 1

```
import pandas as pd

df = pd.read_csv('expt.csv')
df.head()
```

EXECUTED OUTPUT

```
Loan_ID  Gender  Married  Dependents  Education  Self_Employed  \
0  LP001    Male    Yes        0      Graduate        No
1  LP002    Female  No        1      Graduate        No
2  LP003    Male    Yes        2  Not Graduate    Yes
3  LP004    Male    Yes        3+    Graduate        No
4  LP005    Female  No        0      Graduate        Yes

ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0             5000             1500         120             360
1             4200              0         100             360
2             6500            1200         150             360
3             7800            1800         200             360
4             3500             500          90             180

Credit_History  Property_Area  Loan_Status
0              1             Urban           Y
1              1             Rural           Y
2              0          Semiurban           N
3              1             Urban           Y
4              1             Rural           Y
```

PYTHON CODE - CELL 2

```
print(df.head())
print(df.tail())
print(df.info())
print(df.describe())
print(df.columns)
```

EXECUTED OUTPUT

```
Loan_ID  Gender  Married  Dependents  Education  Self_Employed  \
0  LP001    Male    Yes        0      Graduate        No
1  LP002    Female  No        1      Graduate        No
2  LP003    Male    Yes        2  Not Graduate    Yes
3  LP004    Male    Yes        3+    Graduate        No
4  LP005    Female  No        0      Graduate        Yes

ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0             5000             1500         120             360
1             4200              0         100             360
2             6500            1200         150             360
3             7800            1800         200             360
4             3500             500          90             180

Credit_History  Property_Area  Loan_Status
0              1             Urban           Y
1              1             Rural           Y
2              0          Semiurban           N
3              1             Urban           Y
```

```

4      1      Rural      Y
Loan_ID Gender Married Dependents Education Self_Employed \
2 LP003 Male Yes 2 Not Graduate Yes
3 LP004 Male Yes 3+ Graduate No
4 LP005 Female No 0 Graduate Yes
5 LP006 Male No 1 Not Graduate No
6 LP007 Female Yes 2 Graduate No

ApplicantIncome CoapplicantIncome LoanAmount Loan_Amount_Term \
2 6500 1200 150 360
3 7800 1800 200 360
4 3500 500 90 180
5 4600 800 110 360
6 5900 1000 140 360

Credit_History Property_Area Loan_Status
2 0 Semiurban N
3 1 Urban Y
4 1 Rural Y
5 1 Semiurban Y
6 0 Urban N

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 13 columns):
# Column Non-Null Count Dtype
---
0 Loan_ID 7 non-null object
1 Gender 7 non-null object
2 Married 7 non-null object
3 Dependents 7 non-null object
4 Education 7 non-null object
5 Self_Employed 7 non-null object
6 ApplicantIncome 7 non-null int64
7 CoapplicantIncome 7 non-null int64
8 LoanAmount 7 non-null int64
9 Loan_Amount_Term 7 non-null int64
10 Credit_History 7 non-null int64
11 Property_Area 7 non-null object
12 Loan_Status 7 non-null object
dtypes: int64(5), object(8)
memory usage: 860.0+ bytes
None

ApplicantIncome CoapplicantIncome LoanAmount Loan_Amount_Term \
count 7.000000 7.000000 7.000000 7.000000
mean 5357.142857 971.428571 130.000000 334.285714
std 1475.191674 607.492896 37.416574 68.033605
min 3500.000000 0.000000 90.000000 180.000000
25% 4400.000000 650.000000 105.000000 360.000000
50% 5000.000000 1000.000000 120.000000 360.000000
75% 6200.000000 1350.000000 145.000000 360.000000
max 7800.000000 1800.000000 200.000000 360.000000

Credit_History
count 7.000000
mean 0.714286
std 0.487950
min 0.000000
25% 0.500000
50% 1.000000
75% 1.000000
max 1.000000
Index(['Loan_ID', 'Gender', 'Married', 'Dependents', 'Education',
      'Self_Employed', 'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
      'Loan_Amount_Term', 'Credit_History', 'Property_Area', 'Loan_Status'],
      dtype='object')

```

PYTHON CODE - CELL 3

```

categorical_cols = ['Gender', 'Married', 'Dependents', 'Education',
                    'Self_Employed', 'Property_Area']

for col in categorical_cols:
    print(f"\nUnique values in '{col}':")
    print(df[col].unique())

```

EXECUTED OUTPUT

```

Unique values in 'Gender':
['Male' 'Female']

Unique values in 'Married':
['Yes' 'No']

Unique values in 'Dependents':

```

```
['0' '1' '2' '3+']

Unique values in 'Education':
['Graduate' 'Not Graduate']

Unique values in 'Self_Employed':
['No' 'Yes']

Unique values in 'Property_Area':
['Urban' 'Rural' 'Semiurban']
```

PYTHON CODE - CELL 4

```
df['Dependents'] = df['Dependents'].replace('3+', 3)
df['Dependents'] = pd.to_numeric(df['Dependents'],
                                errors='coerce').astype('Int64')

print(df['Dependents'].value_counts(dropna=False))
```

EXECUTED OUTPUT

```
Dependents
0    2
1    2
2    2
3    1
Name: count, dtype: Int64
```

PYTHON CODE - CELL 5

```
cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']

for col in cols:
    print(f"\nStatistics for '{col}':")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

EXECUTED OUTPUT

```
Statistics for 'ApplicantIncome':
Mean: 5357.14
Median: 5000.00
Mode: [3500 4200 4600 5000 5900 6500 7800]
Range: 4300.00
Variance: 2176190.48
Standard Deviation: 1475.19

Statistics for 'CoapplicantIncome':
Mean: 971.43
Median: 1000.00
Mode: [ 0  500  800 1000 1200 1500 1800]
Range: 1800.00
Variance: 369047.62
Standard Deviation: 607.49

Statistics for 'LoanAmount':
Mean: 130.00
Median: 120.00
Mode: [ 90 100 110 120 140 150 200]
Range: 110.00
Variance: 1400.00
Standard Deviation: 37.42
```

RESULT

The dataset structure, categories and numerical distributions were inspected successfully, and the requested descriptive measures were obtained.